

Isaac Rankin

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Education

University of British Columbia , Vancouver, BC, Canada PhD in Statistics Supervisors: Saifuddin Syed and Charles Margossian	2025/09 – Present
University of Oxford , Oxford, United Kingdom MSc in Statistical Sciences (Distinction) Supervisor: Garrett M. Morris	2023/10 – 2024/09
University of Victoria , Victoria, BC, Canada BSc in Mathematics and Statistics (Distinction) 97.52% overall course average, GPA: 9.00/9.00 Supervisor: Julie Zhou	2018/09 – 2023/04

Academic experience

Research Assistant <i>Department of Statistics, University of British Columbia</i> Supervisors: Saifuddin Syed and Charles Margossian	2026/05 – Present
Teaching Assistant <i>Department of Statistics, University of British Columbia</i> STAT 302: Introduction to Probability	2026/05 – 2026/06
Teaching Assistant <i>Department of Statistics, University of British Columbia</i> DSCI 200: Navigating Data: Acquisition, Exploration and Management	2026/01 – 2026/04
Graduate Teaching Assistant <i>Department of Statistics, University of British Columbia</i> Assisted with the evaluations of MSc applications for the 2026 cohort.	2026/01 – 2026/02
Teaching Assistant <i>Department of Statistics, University of British Columbia</i> STAT 251: Elementary Statistics	2025/09 – 2025/12
Research Assistant (NSERC USRA) <i>Department of Mathematics and Statistics, University of Victoria</i> Supervisor: Julie Zhou.	2023/05 – 2023/08
Research Assistant <i>Department of Mathematics and Statistics, University of Victoria</i> Supervisor: Julie Zhou.	2023/01 – 2023/04

Research Assistant (NSERC USRA)

2022/05 – 2022/08

Department of Mathematics and Statistics, University of Victoria

Supervisor: Julie Zhou.

Industry experience

Statistician

2025/01 – Present

Fish, Food, & Allied Workers (Contract)

Built an R-package and performed statistical analyses.

Software Developer

2025/05 – 2025/11

Chinook Business Advisory (Contract)

Built tools to automate business evaluations given scanned financial statements.

Technology Prototype Development Intern

2024/06

Lightricity Ltd, Oxford, United Kingdom

Built and improved tools to assess photovoltaic cells.

Publications

Submitted for review

Generalized Guarantees for Variational Inference in the Presence of Even and Elliptical Symmetry

Charles C. Margossian, [Isaac E. Rankin](#), Lawrence K. Saul

Available at [arXiv:2511.01064](#).

Journal articles

RAD-Median and trimmed mean, new multivariate generalizations of the classical estimators

[Isaac Rankin](#), Julie Zhou

Communications in Statistics-Simulation and Computation, 2025, pages 1-19.

Bayesian and maximin A-optimal designs for spline regression models with unknown knots

[Isaac Rankin](#), Julie Zhou

Statistical Papers, 2023, Volume 65, pages 2011-2032.

Theses

Conditional Diffusion Models for Structure-Based Drug Design with Considerations for Physical Plausibility and a Celecoxib Rediscovery Case Study

[Isaac Rankin](#)

University of Oxford, 2024.

Presentations

Posters

2023 Canadian Statistics Student Conference

Poster: Bayesian and maximin A-optimal designs for spline regression models with unknown knots

Awarded 3rd prize undergraduate poster presentation

Carleton University, Ottawa, Ontario, Canada

Honours and awards

Note: Natural Sciences and Engineering Research Council of Canada will be referred to as NSERC.

Canada Graduate Research Scholarship – Doctoral program (NSERC CGRS D) \$120,000	2026 – 2029
Four-Year Doctoral Fellowship \$90,200 University of British Columbia, Vancouver, BC, Canada	2025 – 2029
Vector Institute Research Grant \$6,000	2026
Youth Community Leadership Award Comox Valley Chamber of Commerce	2025
President’s Academic Excellence Initiative PhD Award \$1,260 University of British Columbia, Vancouver, BC, Canada	2025
Faculty of Science Graduate Entrance Scholarship \$8,500 University of British Columbia, Vancouver, BC, Canada	2025
Third Prize, Undergraduate Poster Presentation Competition Canadian Statistics Student Conference	2023
Undergraduate Student Research Award (NSERC USRA) \$6,000	2023
Constance Van Eeden Scholarship \$3,000 (Declined) University of British Columbia, Vancouver, BC, Canada	2023
Departmental Entrance Scholarship \$2,000 (Declined) University of British Columbia, Vancouver, BC, Canada	2023
Dean’s List (Top 10%) University of Victoria, Victoria, BC, Canada	2020 – 2023
Leeder Family Memorial Award in Mathematics \$3,461 University of Victoria, Victoria, BC, Canada	2022
Columbia Coast Mission Scholarship \$1,166 University of Victoria, Victoria, BC, Canada	2022
Undergraduate Student Research Award (NSERC USRA) \$6,000	2022
President’s Scholarship \$1,498 University of Victoria, Victoria, BC, Canada	2021
Leeder Family Memorial Award in Mathematics \$2,502 University of Victoria, Victoria, BC, Canada	2021
Transfer Scholarship \$2,500 University of Victoria, Victoria, BC, Canada	2020

Dean's Honour Roll (Highest academic standing) Simon Fraser University, Burnaby, BC, Canada	2018 – 2020
President's Honour Roll (High academic standing) Simon Fraser University, Burnaby, BC, Canada	2018 – 2020
Undergraduate Open Scholarship \$2,500 Simon Fraser University, Burnaby, BC, Canada	2020
Alumni Scholarship \$1,250 Simon Fraser University, Burnaby, BC, Canada	2020
Alumni Scholarship \$1,250 Simon Fraser University, Burnaby, BC, Canada	2019
Celebration of Excellence in Mathematics (Top of the class) Mark R. Isfeld Secondary, Courtenay, BC, Canada	2015 – 2018
Celebration of Excellence in Woodwork (Top of the class) Mark R. Isfeld Secondary, Courtenay, BC, Canada	2016

Volunteer activities

Co-president of the Statistics Graduate Student Association (SGSA) Department of Statistics, University of British Columbia, Vancouver, BC, Canada	2025/09 – Present
Co-organizer of the 2026 Dr. Constance van Eeden seminar Guest speaker: Dr. Ryan Tibshirani, University of California, Berkeley Department of Statistics, University of British Columbia, Vancouver, BC, Canada	2025/11 – 2026/04
Statistics workshop teaching assistant Life Sciences Symposium, University of British Columbia, Vancouver, BC, Canada	2026/04/10
General volunteer Filberg Festival, Comox, BC, Canada	2025/08/1-2
Statistician Contributed to the ACE, BE-FIT, and ECHO studies. Cancer Exercise & Physiotherapy Lab, University of British Columbia	2025/05 – 2025/08
Statistician Investigated factors impacting northern resident orcas. Raincoast Conservation Foundation	2025/04 – 2025/06
Ride Leader Mentored high school athletes on the mountain bike team. Mark R. Isfeld Secondary, Courtenay, BC, Canada	2025/03 – 2025/06
Trail Maintenance and Event Volunteer United Riders of Cumberland, Cumberland, BC, Canada	2012 – Present